习题 1 设$u=e^{a_1x_1+\cdots+a_nx_n}$, 其中$\sum\limits_{i=1}^{n}a_i^2=1$. 请验证$\Delta u=0$.

习题 2 验证函数$z=\ln(e^x+e^y)$是方程

$$\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=1 \quad 及 \quad \frac{\partial^2 z}{\partial x^2}\frac{\partial^2 z}{\partial y^2}-\left(\frac{\partial^2 z}{\partial x\partial y}\right)^2=0$$

的解.

习题 3 请按如下要求回答问题:

(1) n摩尔气体的 van der Waals 方程为

$$\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT,$$

其中P为压强, V为体积, T为温度, R为气体常数, a与b为对应该气体的特征常数. 计算 $\partial T/\partial P$ 及 $\partial T/\partial V$;

(2) 适用于固定质量m的理想气体定律方程为$PV=mRT$, 其中P为压强, V为体积, T为温度, R为气体常数, 请验证 $\dfrac{\partial P}{\partial V}\dfrac{\partial V}{\partial T}\dfrac{\partial T}{\partial P}=-1$ 及 $T\dfrac{\partial P}{\partial T}\dfrac{\partial V}{\partial T}=mR$.

习题 4 设$z=f(x,y)$, $x=s+t$, $y=s-t$. 验证 $\dfrac{\partial z}{\partial x}+\dfrac{\partial z}{\partial y}=0$.

习题 5 验证函数$z=f(x+at)+g(x-at)$是如下波动方程 $\dfrac{\partial^2 z}{\partial t^2}=a^2\dfrac{\partial^2 z}{\partial x^2}$.

习题 6 设$u=f(x,y)$, $x=e^s\cos t$, $y=e^s\sin t$. 验证

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=e^{-2s}\left[\frac{\partial^2 u}{\partial s^2}+\frac{\partial^2 u}{\partial t^2}\right].$$

习题 7 设$z=f(x,y)$, $x=r\cos t$, $y=r\sin t$. 验证

$$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=\frac{\partial^2 z}{\partial r^2}+\frac{1}{r^2}\frac{\partial^2 z}{\partial \theta^2}+\frac{1}{r}\frac{\partial z}{\partial r}.$$

习题 8 设$z=f(x,y)$, $x=g(s,t)$, $y=h(s,t)$, 其中各个部分的函数f,g,h均可微. 验证

$$\frac{\partial^2 z}{\partial t^2}=\frac{\partial^2 z}{\partial x^2}\left(\frac{\partial x}{\partial t}\right)^2+2\frac{\partial^2 z}{\partial x\partial y}\frac{\partial x}{\partial t}\frac{\partial y}{\partial t}+\frac{\partial^2 z}{\partial y^2}\left(\frac{\partial y}{\partial t}\right)^2+\frac{\partial z}{\partial x}\frac{\partial^2 x}{\partial t^2}+\frac{\partial z}{\partial y}\frac{\partial^2 y}{\partial t^2}.$$